



AIMS05-01-006

Issue 5

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AIMS
Airbus Material Specification

Carbon Fibre Reinforced Epoxy Prepreg
Twill 2x2 weave style / 180 °C curing class
Standard modulus fibre
Structural material

Material Specification

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1 Scope

This Material Specification defines the requirements of carbon fibre reinforced fabric prepreg epoxy (2x2 Twill weave style), curing at 180 °C, for aerospace application.

The requirements specified in this Material Specification shall not be used as stress allowables.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

This material is used for manufacturing primary airframe and aerospace structures (sandwich and monolithic). The material is to be delivered according to the relevant ABS.

3 Normative References

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS05-01-000 Part 1	Technical Specification - Carbon fibre reinforced epoxy prepreg - General requirements.
AIMS05-01-000 Part 3	Technical Specification – Qualification Programme - Carbon fibre reinforced epoxy prepreg - Qualification and batch release testing fabric (2x2 Twill) / 180 °C curing class - Structural material.
AIMS05-01-000 Part 100	Carbon fiber reinforced epoxy prepreg Structural material Acceptance Testing and Storage Life Prolongation
ABS5003	Semi finished product specification - Carbon fibre reinforced epoxy preregs fabric (twill 2x2 weave style) -180 °C-curing class - Structural material.
AITM1.0001	Fiber reinforced plastics - Determination of mechanical degradation due to chemical paint strippers.
AITM1.0002	Fiber reinforced plastics - Determination of in-plane shear properties ($\pm 45^\circ$ tensile tests).
AITM1.0003	Determination of the glass transition temperature.
AITM1.0005	Carbon fiber reinforced plastics materials - Determination of interlaminar fracture toughness energy – mode I.
AITM1.0006	Carbon fiber reinforced plastics materials - Determination of interlaminar fracture toughness energy – mode II.
AITM1.0010	Fiber reinforced plastics - Determination of compression strength after impact.
AITM3.0001	Analysis of thermoset systems by High performance liquid chromatographie.
AITM3.0002	Analysis of non-metallic materials (uncured) by Differential scanning calorimetry.
AITM3.0003	Analysis of organic compounds by Infrared spectroscopy.
AITM3.0004	Determination of gel time and viscosity.
AITM3.0008	Determination of the extent of cure by Differential Scanning Calorimetry.
EN2155-12	Aerospace series - Test method for the determination of the linear thermal expansion.
EN2378	Aerospace series - Determination of water absorption by immersion in demineralized water.
EN2557	Aerospace series - Carbon fiber preimpregnates - Test method for the determination of mass per unit area.
EN2558	Aerospace series - Carbon fiber preimpregnates - Test method for the determination of the percentage of volatile matter.
EN2559	Aerospace series - Carbon fiber preimpregnates - Test method for the determination of the resin and the fiber content and the mass of fiber per unit area.
EN2560	Aerospace series - Carbon fiber preimpregnates - Test method for the determination of the resin flow.

EN2561	Aerospace series - Carbon thermosetting resin unidirectional laminates - Tensile test parallel to the fiber direction.
EN2563	Aerospace series - Unidirectional laminates carbon thermosetting resin - Determination of interlaminar shear strength.
EN2850	Aerospace series - Fiber reinforced plastics - Testing unidirectional laminates and woven fabric laminates - Compression test.
EN3615	Aerospace series, Fiber reinforced plastics - Determination of the conditions of exposure to humid atmosphere and the determination of moisture absorption.
ISO10119	Carbon fiber, determination of density.
ISO1183 method A	Plastics, methods for determining the density and relative density of non-cellular plastics.

4 Technical requirements

4.1 General

Specification values are minimum except where stated.

4.1.1 Material description

See table 1.

Table 1: Material description

Characteristic		Material description	
Material type		Carbon fiber fabric/epoxy resin	
Formulation	Fibre	Carbon, standard modulus	
	Fabric	Fabric with balanced weave. Twill 2x2	
	Matrix	Epoxy, toughened, controlled flow	
Form and method of production		Fabric, cured in an autoclave at 180 °C	
Resin content by weight, uncured		Type A (mF=285 g/m ²) 42 % nominal	Type B (mF=285 g/m ²) 40 % nominal

4.1.2 Worklife and storage life

See table 2.

Table 2: Worklife and storage conditions

Properties	Shop conditions	Unit	Requirements
Storage life -18 °C	-	months	> 12
Worklife	30 °C/75 % RH ¹⁾	days	> 10
	25 °C ± 2 / 45 % RH ± 10	days	> 10

¹⁾ The prepreg can be exposed to an additional minimum of 5 days at the above mentioned shop condition provided that the lay-up is fully bagged and ready for use

4.1.3 Processing

The curing cycle shall be agreed between the manufacturer and Airbus at the onset of the material qualification program and shall be used to produce all test specimens for qualification testing. The laminates for test purposes shall be cured in an autoclave. The curing cycle shall comply with the general requirements reported in table 3. The agreed curing cycle shall be specified in the relevant IPS and shall be used for the manufacture of all the batch release test specimens.

Table 3: Processing conditions

Cure parameter	Unit	Requirements
Cure type: autoclave		
Heating rate	°C/min	0,2 - 5
Cure temperature	°C	180 +10/-5
Cure time	min	120 - 240
Cure pressure	MPa	0,30 - 1,1
Cooling rate	°C/min	0,2 – 3,5
Vacuum (abs)	MPa	0,005 – 0,1
Note: The use of modified cure cycles is permissible if agreed between Airbus and the Supplier, and specified in the QTP or proven suitable by subsequent tests		

4.2 Specific

4.2.1 Physical-chemical properties

Uncured and cured properties see table 4.

4.2.2 Mechanical properties

See table 5.

5 Designation

Refer to ABS5003.

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS05-01-000 Part 1 and Part 3.

7 Incoming inspection

Materials qualified against this AIMS shall undergo “Technical Incoming Inspection” as per AIMS05-01-000 Part 100 prior to use.

Table 4: Physical-chemical properties

Properties		Test methods	Unit	Requirements	
Material code				AIMS05-01-006	
				A	B
Uncured					
Prepreg mass/unit area		EN2557C	g/m ²	490 ± 30	475 ± 25
Fibre mass/unit area		EN2559C	g/m ²	285 ± 10	
Resin density		ISO1183 Method A	g/cm ³	1,29 ± 0,03	
Fibre density		ISO10119	g/cm ³	1,80 ± 0,05	
Volatile content		EN2558	%	< 2	
Resin content		EN2559	%	42 ± 2	40 ± 2
Physico-chemical tests	HPLC	AITM3-0001	-	1)	
	DSC	AITM3-0002	-	1)	
	IR	AITM3-0003	-	1)	
	Gel Time	AITM3-0004 method B or E	-	1)	
	Viscosity	AITM3-0004 Method A	-	1)	
Resin flow		EN2560	%	1)	
Tack		To be agreed between manufacturer and Airbus	-	1)	
Cured					
Coefficient of thermal expansion		EN2155-1 2)	-	3)	
Water uptake		EN3615, 70 °C/85 % R.H.	%	< 1,4	
Water uptake		EN2378, 70 °C/water	%	3)	
Extent of cure		AITM3.0008	%	3)	
Cured ply thickness		acc. to AIMS05-01-000 Part 1	mm	0,285	
Note: The stated requirements are average values					
1) Equipment settings and value to be agreed between manufacturer and Airbus					
2) It is also allowed to use ASTM E831					
3) Results to be reported					

Table 5: Mechanical properties

Properties	Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirement 1)			
					X	Xmin	X	Xmin
Material Code					AIMS05-01-006			
					A		B	
Glass transition temperature, Tg onset	AITM1.0003	°C	dry	-	-	150	-	150
			wet 1	-	-	110	-	110
Tensile strength 0° (warp)	EN2597 type B	MPa	dry	-55	600	550	600	550
				RT	700	620	700	620
				70	750	650	750	650
			wet 1	70	750	650	750	650
				90	750	650	750	650
Tensile modulus 0° (warp)	EN2597 type B	GPa	dry	-55	65 ± 10		65 ± 10	
				RT	65 ± 10		65 ± 10	
				70	65 ± 10		65 ± 10	
			wet 1	70	65 ± 10		65 ± 10	
				90	65 ± 10		65 ± 10	
Poissons Ratio 0/90°	EN2597 type B	-	dry	RT	-	-	-	-
			wet 1	70	-	-	-	-
				90	-	-	-	-
Tensile strength 90° (weft)	EN2597 type B	MPa	dry	-55	650	500	650	500
				RT	700	550	700	550
			wet 1	70	700	550	700	550
				90	700	550	700	550
Tensile modulus 90° (weft)	EN2597 type B	GPa	dry	-55	65 ± 10		65 ± 10	
				RT	65 ± 10		65 ± 10	
			wet 1	70	65 ± 10		65 ± 10	
				90	65 ± 10		65 ± 10	
Compression strength 0° (warp)	EN2850 type B	MPa	dry	-55	800	750	800	750
				RT	740	640	740	640
				70	650	540	650	540
			wet 1	70	630	520	630	520
				90	600	480	600	480
Compression modulus 0° (warp)	EN2850 type B	GPa	dry	-55	58 ± 5		58 ± 5	
			wet 1	RT	58 ± 5		58 ± 5	
				70	58 ± 5		58 ± 5	
				70	58 ± 5		58 ± 5	
				90	58 ± 5		58 ± 5	
Compression strength 90° (weft)	EN2850 type B	MPa	dry	RT	740	620	740	620
			wet 1	70	610	500	610	500
				90	590	460	590	460
Compression modulus 90° (weft)	EN2850 type B	GPa	dry	RT	58±5		58±5	
			wet 1	70	58±5		58±5	
				90	58±5		58±5	
(continued)								

(continued)

Table 5: Mechanical properties

Properties	Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirement 1)			
					X	Xmin	X	Xmin
Material Code					AIMS05-01-006			
					A		B	
In plane shear strength	AITM1.0002	MPa	dry	-55	110	105	110	105
				RT	100	90	100	90
				70	90	80	90	80
				120	70	60	70	60
			wet 1	RT	100	85	100	85
				70	80	70	80	70
				90	70	65	70	65
			120	55	50	55	50	
In plane shear modulus	AITM1.0002	GPa	dry	-55	5 ± 2		5 ± 2	
				RT	5 ± 2		5 ± 2	
				70	5 ± 2		5 ± 2	
				120	4 ± 2		4 ± 2	
			wet 1	RT	5 ± 2		5 ± 2	
				70	4 ± 2		4 ± 2	
				90	4 ± 2		4 ± 2	
			120	3 ± 1,5		3 ± 1,5		
ILSS 0°	EN2563	MPa	dry	-55	60	55	60	55
				RT	60	55	60	55
				70	60	55	60	55
				120	45	40	45	40
			wet 1	RT	60	55	60	55
				70	45	40	45	40
				90	40	38	40	38
				120	35	30	35	30
wet 2	70	40	38	40	38			
	Compression after impact 4)	AITM1.0010	MPa	dry	RT	-	-	-
			wet 1	70	-	-	-	-
G1C	AITM1.0005	J/m²	dry	RT	695	650	695	650
			wet 1	70	650	600	650	600
G2C	AITM1.0006	J/m²	dry	RT	-	-	-	-
			wet 1	70	-	-	-	-
Resistance to aggressive fluids (ILSS) 3)								
Reference	EN2563	MPa	dry	RT	65	60	65	60
			wet 1	70	45	40	45	40
Skydrol	EN2563	MPa	70°C/1000 h	70	5)	5)	5)	5)
Skydrol/water	EN2563	MPa	70°C/1000 h	70	5)	5)	5)	5)
Fuel	EN2563	MPa	RT/1000 h	70	5)	5)	5)	5)
MEK	EN2563	MPa	RT/1 h	RT	5)	5)	5)	5)
Resistance to paint strippers 4)								
Phenolic	AITM1-0001	MPa	-	RT	5)	5)	5)	5)
Non-phenolic	AITM1-0001	MPa	-	RT	5)	5)	5)	5)
(continued)								

(continued)

Table 5: Mechanical properties (concluded)

Properties	Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirement 1)			
					X	Xmin	X	Xmin
Material Code					AIMS05-01-006			
					A		B	
Plain tensile strength	AITM1.0007A 25/50/25	MPa	dry	-55	500	430	500	430
				RT	550	470	550	470
			wet 1	70	545	470	545	470
				70	545	470	545	470
Open Hole tensile strength	AITM1.0007B 25/50/25	MPa	dry	-55	280	240	280	240
				RT	280	255	280	255
			wet 1	70	290	260	290	260
Plain compression strength	AITM1.0008A 25/50/25	MPa	dry	RT	450	400	450	400
				70	500	430	500	430
			wet 1	70	380	330	380	330
Open Hole compression strength	AITM1.0008B 25/50/25	MPa	dry	-55	300	280	300	280
				RT	320	280	320	280
			wet 1	70	260	240	260	240
Filled Hole tensile strength	AITM1.0007D 25/50/25	MPa	dry	RT	260	230	260	230
				70	270	230	270	230
			wet 1	90	270	230	270	230
Filled Hole compression strength	AITM1.0008D 25/50/25	MPa	dry	-55	540	450	540	450
				RT	440	380	440	380
				70	360	320	360	320
			wet 1	90	400	300	400	300
Bearing strength 0,5%	AITM1.0009A 25/50/25	MPa	dry	-55	720	620	720	620
				RT	650	610	650	610
			wet 1	70	550	480	550	480
				120	490	450	490	450
Bearing strength 2%	AITM1.0009A 25/50/25	MPa	dry	-55	810	750	810	750
				RT	750	730	750	730
			wet 1	70	640	610	640	610
				120	560	510	560	510
Bearing strength 6%	AITM1.0009A 25/50/25	MPa	dry	-55	890	790	890	790
				RT	840	810	840	810
			wet 1	70	660	630	660	630
				120	540	510	540	510
Bearing strength Ultimate	AITM1.0009A 25/50/25	MPa	dry	-55	950	900	950	900
				RT	880	860	880	860
			wet 1	70	690	660	690	660
				120	590	560	590	560

- 1) Requirement for Tensile, Compression, Plain, Open hole and filled hole Tensile and Compression have been calculated using a theoretical thickness per ply of 0,285 mm. Requirements calculated with gross section. Other requirements are calculated with the real thickness.
X means minimum average values.
Xmin means minimum individual values.
- 2) DRY: According to EN2743; WET 1: According to EN2823; WET 2: Water exposure in compliance to EN2378
- 3) EN2563 plus AIMS05-01-000 Part1 para. 4.2.3.3.
- 4) AIMS05-01-000 Part1 para. 4.2.3.4.
- 5) The values after conditioning will not have a decrease greater than the corresponding one with the ILSS or In-plane shear reference value at equivalent test temperature.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1		New Standard.
2		
3 05/05		Revision of requirement.
4 05/14	Chapter 3 Chapter 7	AIMS05-01-000 part 100 added Chapter 7 defined in line with Incoming Inspection (Part 100) implementation
5 03/17		Revision of requirements Table 5: Mechanical properties (for both Grades A & B): - In plane shear strength (dry/-55°C) - In plane shear strength (wet/120°C) - Open Hole compression strength (wet/70°C) - Bearing strengths (dry and wet) 0.5%, 2%, 6% and Ultimate